HackIIITD

- **TEAM NAME:** D-CRYPT
- **THEME:** Blockchain and Cybersecurity
- **PROBLEM STATEMENT TITLE:** Automated Attribution of Unknown Cryptocurrency Wallets to Nearest Virtual Asset Service Providers (VASPs) through Blockchain Intelligence APIs
- **PS CATEGORY:** Software
- **TEAM MEMBERS:**

i) DYUTIRANJAN PATRA [LEADER]
ii) JITESH PRASHAD
iii) TINA SAHA

Powered by TinyFish

D-CRYPT SHIELD

India's Blockchain Intelligence & VASP Attribution Engine

❑ The Challenge

Suspect wallets are known but the VASP receiving the funds often isn't. Multi-hop, cross-chain flows make manual attribution slow and delay lawful action.

❑ Our Approach

D-CRYPT SHIELD automates multi-chain tracing, probable VASP attribution, confidence scoring, and evidence-ready lawful action.

❑ Key Innovations

➢ VASP Attribution

BFS hop distance + direct-deposit verification + tagged VASP addresses + cluster connectivity

➢ Graph Intelligence

Visual reconstruction of wallet-to-wallet fund flows

➢ Live watch & alerting

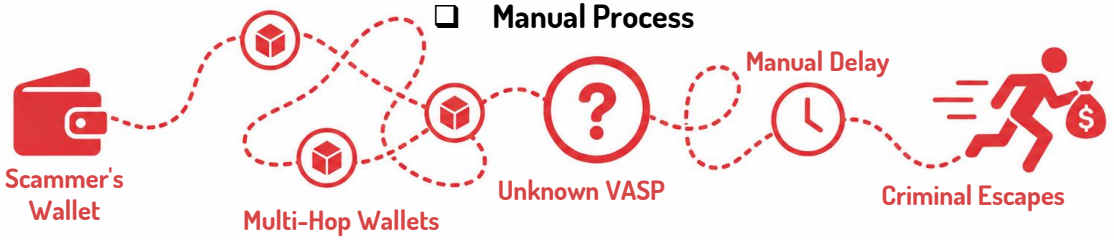
Mempool monitoring → real-time officer alerts

➢ Civilian Reporting Portal

(OTP/ID Verified) citizen reports + Autonomous Web Forensics (TinyFish)

❑ PROBLEM	(M A P P I N G) →	SOLUTION
Multi-Hop Cross-Chain Tracing Funds move through intermediary wallets, bridges, mixers and multiple chains.		**Omni-Chain BFS Engine** Hop-by-hop tracing across BTC, ETH, TRON, SOL, Polygon, BNB & more.
Raw Blockchain Data Thousands of transactions and addresses are difficult to interpret.		**Visual Intelligence** Interactive graph visualization reconstructs wallet-to-wallet fund movement.
Unknown VASP The suspect wallet is known, but its receiving VASP may be unknown.		**VASP attribution & Confidence Scoring** Probable VASP attribution using hop-count, deposit patterns, timing and wallet connectivity.
Delayed Action Manual tracing and evidence preparation delay lawful requests.		**Automation + Sahyog API integration** Evidence-ready reports and SAHYOG-aligned routing for lawful disclosure/freezing workflows.

❑ Manual Process

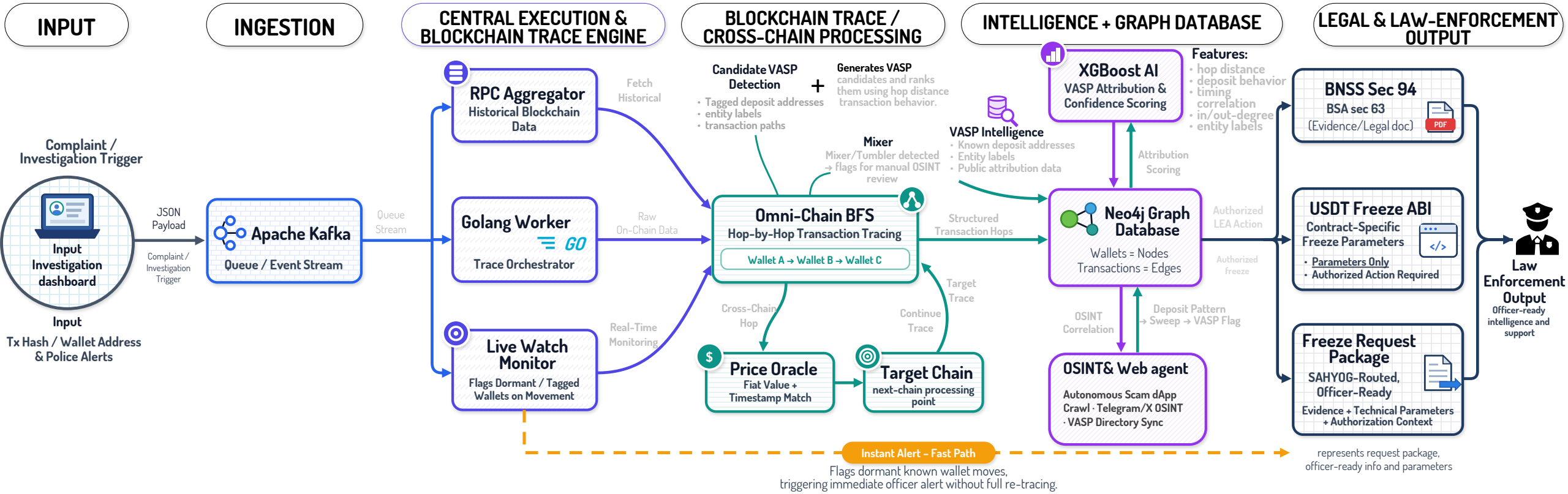


❑ D-CRYPT SHIELD



TECHNICAL APPROACH

❑ METHODOLOGY



❑ Backend Core

Golang · Geth/Ethers.js · Custom RPC Aggregator

❑ Intelligence Service

TinyFish

Python · XGBoost · TinyFish API · OSINT Scrapers · ReportLab/PyQRCode

❑ Frontend Dashboard

Next.js · React · Tailwind CSS · D3.js

❑ Database Engine

Neo4j Graph · PostgreSQL

❑ DevOps & Streaming

Docker · Kubernetes · Apache Kafka

HOW WE PLAN TO INTEGRATE THE TinyFish API [P-1]

The Forensic Integration Thesis: **D-CRYPT** SHIELD reconstructs on-chain fund flows via Omni-Chain BFS. However, **90% of cryptocurrency frauds originate off-chain** on phishing websites, fake task-investment dApps and cloned exchange portals. We integrate **TinyFish API as our autonomous Web Operations & Agentic OSINT Layer** to preserve ephemeral web evidence before domain takedowns, automate legal notice submissions to exchange portals, and keep our VASP nodal directory synchronized.

Core API Primitives & Endpoints Used

`POST /agent`	Goal-driven vision model (FIRE-1) for autonomous web portal navigation and form filling.
`POST/fetch &/scrape`	Headless Chromium rendering for dynamic Web3 React/Next.js pages.
`GET /search`	Structured, rank-stable JSON search engine for darknet & scam forum OSINT.
`Browser API (CDP)`	Remote Chrome DevTools Protocol session over WebSockets via residential proxies.

Operational Reliability & Anti-Bot Infrastructure

Cloudflare / CAPTCHA Bypass
Managed residential proxy rotation and canvas fingerprint spoofing.

Client-Side Hydration
Waits for full DOM hydration to unhide dynamic Web3 wallet-connect modals.

Evidence Preservation Speed
Completes autonomous page captures in < 8 seconds before domain takedowns.

INTEGRATION PIPELINE

CASE TRIGGER
- Citizen scam URL or officer case trigger
- Victim reports link

→ **TinyFish LAYER**
- Headless Chromium execution
- Cloudflare & anti-bot bypass
- Vision-model form navigation

→ **EXTRACTED OFF-CHAIN INTEL**
- Timestamped DOM evidence
- Embedded deposit wallets
- Exchange filing ticket ID

→ **D-CRYPT SHIELD ENGINE**
- Omni-Chain BFS hop tracing
- Neo4j graph attribution
- BNSS Sec 94 frozen dossier

KEY METRICS

< sec	Evidence capture, pre-takedown
min → sec	Exchange LEA filing time
Residential proxy	Cloudflare & anti-bot bypass

HOW WE PLAN TO INTEGRATE THE TinyFish API [P-2]

Module 1: Ephemeral Scam & Phishing Forensics

Integrated into: /civilian Intake Portal

THE PROBLEM Scammers take down fake investment dApps and review portals within 24–48 hours, destroying evidence before police register an FIR.

TINYFISH IMPLEMENTATION

- Victim reports a scam URL → TinyFish instantly triggers a stealth browser run.
- Residential proxy routing + anti-bot stealth bypasses Cloudflare and CAPTCHAs.
- Executes client-side React/Web3 JavaScript to render hidden wallet-connect modals.
- Captures tamper-evident full-page DOM screenshot; extracts deposit wallets & contract hashes.

OUTCOME Zero evidence loss; extracted wallets automatically seed our BFS Trace Engine.

Module 2: Autonomous Exchange LEA Portal Filing

Integrated into: /reports Legal Output Module

THE PROBLEM Officers lose 30–45 min per case logging into exchange LEA desks (e.g., Binance Kodex, CoinDCX LEA Portal) to submit freeze requests.

TINYFISH IMPLEMENTATION

- Goal-Driven Autonomous Agent interacts with authorized exchange LEA portals.
- Navigates multi-step forms, enters suspect wallets, uploads signed BNSS Sec 94 / Sec 91 notice PDF and submits.
- Scrapes the Filing Acknowledgement Ticket Number into the officer's active case dossier.

OUTCOME 90% faster emergency freezing; manual portal data entry becomes 1-click execution.

Module 3: Real-Time FIU-IND VASP Directory Sync

Integrated into: /lookup Compliance Database

THE PROBLEM Exchanges frequently change legal entities, nodal compliance officer emails and LEA grievance desks, causing bounced statutory notices.

TINYFISH IMPLEMENTATION

- Scheduled agent crawls FIU-IND gazettes, exchange terms and compliance pages.
- Extracts structured JSON: Nodal Officer names, LEA emergency emails, registered addresses.
- D-CRYPT SHIELD's VASP database updates continuously with zero manual curation.

OUTCOME 100% notice deliverability; no bounce-backs during critical fund-freezing windows.

FEASIBILITY, SCALABILITY AND VIABILITY

❑ Data Feasibility

Public blockchain ledgers can be accessed through standard RPC nodes.

❑ Financial Feasibility

Open-source core avoids recurring blockchain intelligence Costs compared with commercial platforms

❑ Legal Feasibility

Designed to support lawful disclosure and authorized freezing workflows with evidence provenance &cryptographic integrity checks.

❑ Operational Feasibility

API-ready architecture for SAHYOG integration, case-based LEA workflows, and officer-ready evidence packaging.

❑ Potential Challenge	❑ Strategy for Overcoming It
RPC Rate Limiting Large-scale tracing can hit provider rate limits.	**RPC Aggregator** Async queueing + provider failover + rate-limit-aware workers.
Graph Database scale Large multi-hop graphs increase memory and query load.	**K8s & Query Pagination** Query pagination + bounded graph traversal + horizontal scaling
Data Sovereignty Sensitive investigations may require controlled infrastructure.	**Air-Gapped Deployment** Self-hosted nodes + controlled archive infrastructure.
Evidence Integrity Automated reports require verifiable provenance.	**Integrity Controls** Cryptographic hashes + timestamps + provenance metadata.
Stagnant / Offshore Funds Funds may remain inactive, or foreign VASPs may not respond.	**Jurisdiction Aware Packaging** Evidence + VASP-specific request data for authorized disclosure/freezing workflows.

❑ Technical Feasibility

01 REAL-TIME BLOCKCHAIN MONITORING

Public blockchain data is already available.

Blockchain Networks

(via public nodes / RPC / WebSocket)

❑ Live On-Chain Data

- Transactions
- Pending TXs
- Smart Contract Events
- Block Data

02 ON-CHAIN ACTIVITY DETECTION

Decode transaction/events to identify fund movements.

Smart Contract Event Decoding

- Token Swap
- Token Transfer
- Amount & Tokens
- DEX / Protocol
- Timestamp
- Transaction Hash

03 AUTOMATED VASP ATTRIBUTION

Match wallet addresses with known entities using labelled datasets and VASP registries.

Labelled Data Sources

- VASP Registries
- Exchange Disclosures
- On-chain Labels (Etherscan, Chainalysis, etc.)
- Heuristic Clustering

VASP Identified

e.g. Binance, Coinbase, WazirX, OKX

Entity	Binance
Cluster Type	Exchange
Confidence	High
Category	VASP

IMPACT AND BENEFITS

❑ TARGET USERS

✓ **Civilians (Victims)**
Report suspected wallets & scam dApp URLs with automated web evidence preservation.

✓ **State Police (LEAs)**
Automates repetitive tracing and supports faster identification of probable VASPs.

✓ **MHA & Government**
Supports controlled government deployment with scalable multi-chain analysis and case-based intelligence.

❑ LEGAL AND INVESTIGATIVE VALUE

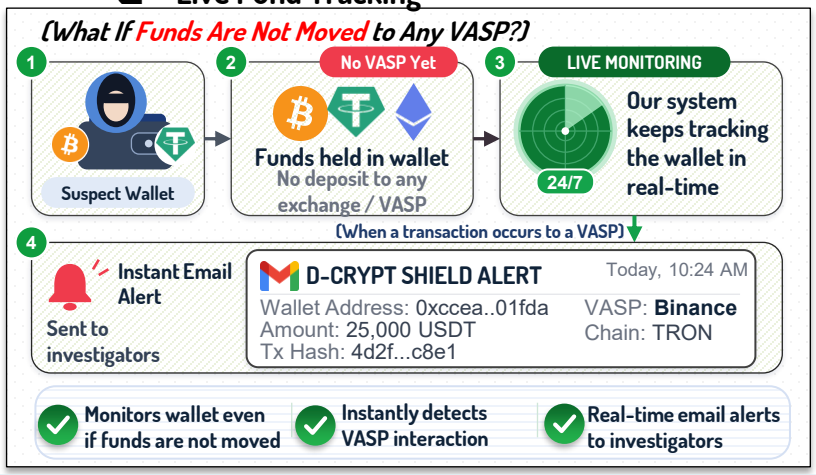
✓ **Evidence Ready Investigation**
Evidence-ready reports with cryptographic hashes, timestamps and provenance metadata.

✓ **Jurisdiction Aware Request**
Generates jurisdiction-aware data for authorized disclosure/freezing workflows.

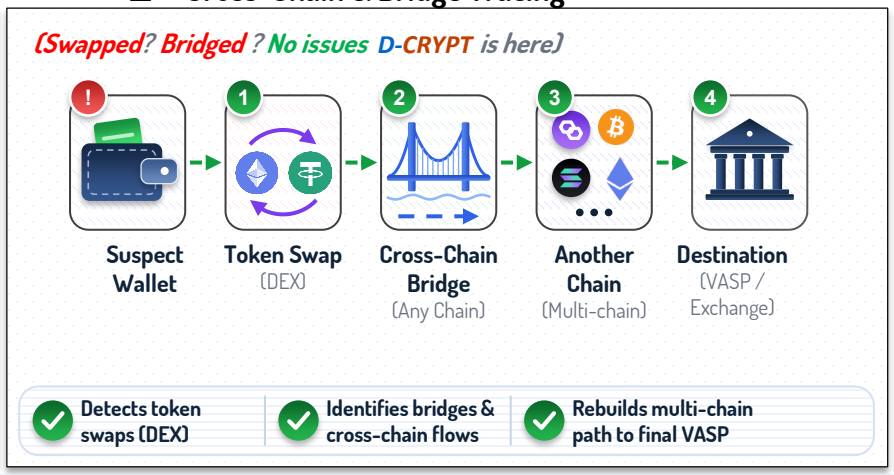
✓ **Investigator Assisted Intelligence**
Reduces manual tracing effort so LEAs can focus on actionable investigative intelligence.

❑ CORE FORENSIC USE CASES

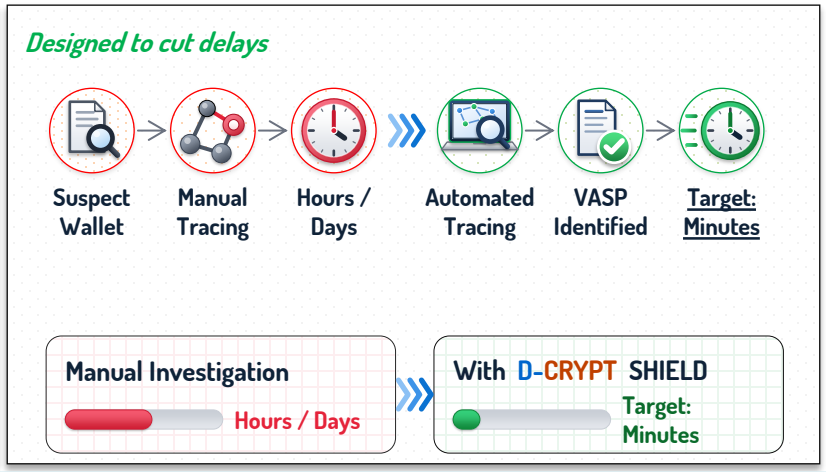
❑ Live Fund Tracking



❑ Cross-Chain & Bridge Tracing



❑ Faster VASP Attribution



RESEARCH AND REFERENCES

❑ GROUND-TRUTH DATASETS & REGISTRIES

- **FIU-IND Registered VASP Directory:**
  Official registry of 28+ compliant Indian exchanges (CoinDCX, WazirX, CoinSwitch, ZebPay) with mapped nodal compliance officers.
- **Public Multi-Chain Entity Labels (Etherscan, Tronscan):**
  Curated database of over 100,000+ verified exchange deposit gateways, cold storage vaults, and hot wallets across 6 blockchains.
- **OFAC SDN & Illicit Mixer Blacklists:**
  US Treasury & international sanction lists for flagging mixers (Tornado Cash, Blender, Sinbad) and state-sponsored cybercrime addresses.
- **TinyFish AI Web Operations Framework:**
  Standardized agentic browsing protocol (agent.tinyfish.ai) for ephemeral scam portal forensics and dynamic VASP compliance page sync.

❑ TECHNICAL & LEGAL EXECUTION PROOFS

- **Pre-Emptive Tracking**
  Official Web3.js docs for eth.subscribe('pendingTransactions').
- **AI Graph Science**
  Neo4j academic docs for Heuristic Graph Clustering in Financial Fraud.
- **Legal Evidence**
  BNSS (Sec 94) & Indian Evidence Act (Sec 63) legal frameworks.
- **VASP & Wallet Address Attribution Data**
  Eth Labels — Public Multi-Chain Blockchain Address Labels

❑ REAL CRYPTO FRAUD CASES

01 — Nov 2025

₹285 CRORE
CYBER-FRAUD PROCEEDS

- Collected through **30+** bank accounts
- Converted to cryptocurrency
- USDT purchased via **Binance P2P**
- **₹4.81 Cr** USDT converted via **CoinDCX**

Bank Accounts → USDT (P2P) → Binance → CoinDCX

Source: ED Press Release (Cyber Fraud Case, 20 Nov 2025)

02

₹90 CRORE
CRYPTO ASSETS FROZEN

E-Nuggets online gaming scam

Crypto assets across **70 accounts**

03 — 2024 – 2025

₹1,646 CRORE
BITCOIN SEIZED

- Crypto-investment fraud
- Proceeds layered through multiple crypto wallets
- Bitcoin valued at **₹1,646 crore** seized
- **₹489 crore** assets attached in the case

Multiple Crypto Wallets → Layering & Mixing → Complex Money Trail

Source: ED Annual Report 2024–25 (Case Study 4)

❑ PROBLEM CONTEXT & EVIDENCE

Official data for year-wise submitted cases · Source: ED
Link: Click [PAGE-38, Sr.No. 3]

